

All of Statistics: A Concise Course in Statistical Inference
Solutions by David A. Lee

All errors, typographical and substantive, and other offenses, are entirely my own.

Chapter 2: Random Variables

Import Packages

Please see the associated GitHub repo for all code and comments to simulations.
[LINK HERE]

```
import numpy as np
from numpy.random import choice
import random
import matplotlib.pyplot as plt
import scienceplots
```

Question: 2.14.1

Show that

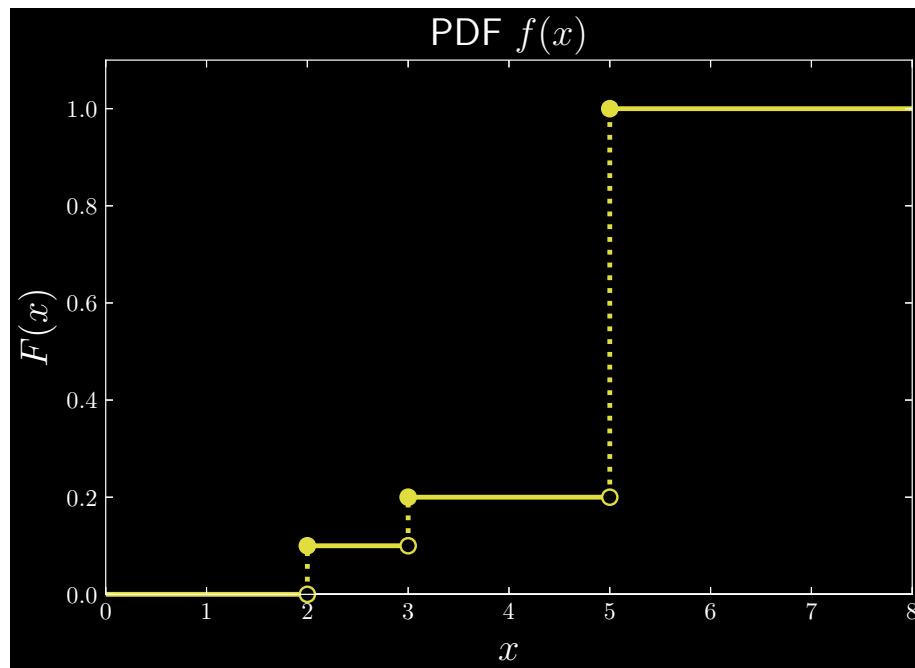
$$\mathbb{P}(X = x) = F(x^+) - F(x^-).$$

PROOF. By Lemma 2.15(1), $\mathbb{P}(X = x) = F(x) - F(x^-)$ where $F(x^-) = \lim_{y \uparrow x} F(y)$. By Theorem 2.8(iii), F is right-continuous, or $F(x) = F(x^+)$, where $F(x^+) = \lim_{y \rightarrow x} F(y)$. Thus $\mathbb{P}(X = x) = F(x^+) - F(x^-)$. $\square$

Question: 2.14.2

Let X be such that $\mathbb{P}(X = 2) = \mathbb{P}(X = 3) = 1/10$ and $\mathbb{P}(X = 5) = 8/10$. Plot the CDF F . Use F to find $\mathbb{P}(2 < X \leq 4.8)$ and $\mathbb{P}(2 \leq X \leq 4.8)$.

The CDF is given by



And the respective probabilities are

$$\begin{aligned}
 \mathbb{P}(2 < X \leq 4.8) &= \mathbb{P}(X \leq 4.8) - \mathbb{P}(X \leq 2) \\
 &= 2/10 - 1/10 \\
 &= \boxed{1/10} \\
 \mathbb{P}(2 \leq X \leq 4.8) &= \mathbb{P}(X \leq 4.8) - \mathbb{P}(X < 2) \\
 &= 2/10 - 0 \\
 &= \boxed{1/5}
 \end{aligned}$$

Question: 2.14.3

Prove Lemma 2.15.

Lemma (Wasserman 2.15)

Let F be the CDF for a random variable X . Then:

1. $\mathbb{P}(X = x) = F(x) - F(x^-)$ where $F(x^-) = \lim_{y \uparrow x} F(y)$;
2. $\mathbb{P}(x < X \leq y) = F(y) - F(x)$;
3. $\mathbb{P}(X > x) = 1 - F(x)$;
4. If X is continuous then

$$\begin{aligned}
 F(b) - F(a) &= \mathbb{P}(a < X < b) = \mathbb{P}(a \leq X < b) \\
 &= \mathbb{P}(a < X \leq b) = \mathbb{P}(a \leq X \leq b).
 \end{aligned}$$

PROOF. 1. Let $F(x^-) = \lim_{y \uparrow x} F(y)$. Then let $y_1, y_2, \dots$ be a sequence such that $y_1 < y_2 < \dots$ and $\lim_i y_i = x$. Then let $A_i = (-\infty, y_i]$ and $A = (-\infty, x]$, so $A = \bigcup_{i=1}^{\infty} A_i$ and $A_1 \subset A_2 \subset \dots$. Thus by continuity of probability, $\lim_i \mathbb{P}(A_i) = \mathbb{P}(\bigcup_i A_i) = \mathbb{P}(A) = F(x)$. But $\lim_i \mathbb{P}(A_i) = F(x^-)$ by definition, so we have $F(x) = F(x^-)$, implying $\mathbb{P}(X = x) = F(x) - F(x^-) = 0$.

2. We have $\mathbb{P}(X \leq y) = F(y)$ and $\mathbb{P}(X \leq x) = F(x)$. Using mutual exclusivity of the events $\{X: X \leq x\}$ and $\{X: x < X \leq y\}$, we can conclude that $\mathbb{P}(X \leq x) + \mathbb{P}(x < X \leq y) = \mathbb{P}(X \leq y)$ implies $\mathbb{P}(x < X \leq y) = F(y) - F(x)$.

3. By mutual exclusivity of the events $\{X: X > x\}$ and $\mathbb{P}(X \leq x)$, we must have $\mathbb{P}(X \leq x) + \mathbb{P}(X > x) = 1$, which implies $\mathbb{P}(X > x) = 1 - \mathbb{P}(X \leq x) = 1 - F(x)$.

4. Assume X is continuous. Then $\mathbb{P}(X = a) = \mathbb{P}(X = b) = 0$. Since the events $\{X = a\}, \{X: a < X < b\}, \{X = b\}$ are all disjoint, the desired equalities can be derived from summing the probabilities of the corresponding mutually exclusive events. $\square$

Question: 2.14.4

Let X have the probability density function

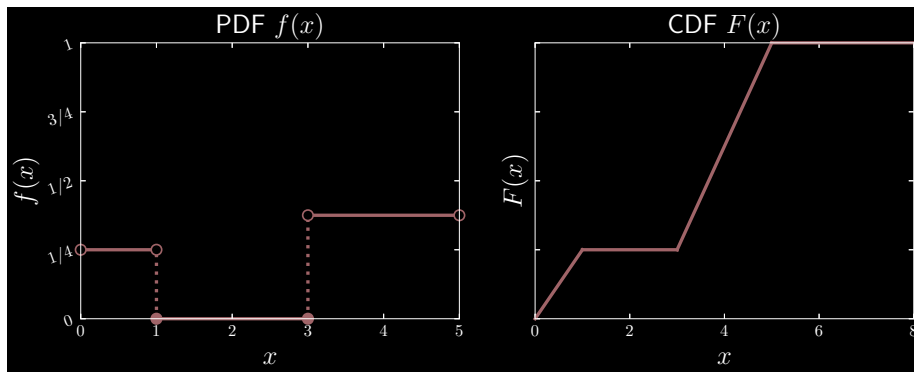
$$f_X(x) = \begin{cases} 1/4 & 0 < x < 1 \\ 3/8 & 3 < x < 5 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the cumulative distribution function of X .
- (b) Let $Y = 1/X$. Find the probability density function $f_Y(y)$ for Y .
Hint: Consider three cases $\frac{1}{5} \leq y \leq \frac{1}{3}$, $\frac{1}{3} \leq y \leq 1$, and $y \geq 1$.

(a) To find the CDF, we find how the area under the PDF changes as x increases. The direct approach is to integrate each segment of the piecewise PDF, with some nuance – as x moves between 1 and 3, since the PDF is zero here, there is no contribution to the CDF, and so $F(x)$ stays at $1/4$ on this segment. Additionally, once we reach the $(3, 5)$ segment, we not only integrate $3/8$ over the aforementioned interval, but add $1/4$ to reflect the *cumulative* probability. Proceeding in this manner, the CDF is

$$F_X(x) = \begin{cases} 0 & x \leq 0 \\ x/4 & 0 < x < 1 \\ 1/4 & 1 \leq x \leq 3 \\ 1/4 + 3(x-3)/8 & 3 < x < 5 \\ 1 & 5 \leq x \end{cases}$$

Graphing the PDF and CDF:



- (b) To derive $F_Y(y)$, we begin with

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leq y) \\ &= \mathbb{P}(1/X \leq y) \\ &= \mathbb{P}(1/y \leq X) \\ &= 1 - \mathbb{P}(X < 1/y) \end{aligned}$$

Since $f_X(x)$ is given as a piecewise function, we must calculate the segments of the CDF $F_Y(y)$, and consequently the PDF $f_Y(y)$, separately.

Note also that

$$\frac{d}{dy}F_Y(y) = f_Y(y)$$

Case 1: $1/5 < y < 1/3$ corresponding to $3 < x < 5$

$$\begin{aligned} F_Y(y) &= 1 - \mathbb{P}(X < 1/y) \\ &= 1 - \left(\frac{1}{4} + \frac{3(1/y - 3)}{8} \right) \\ &= \frac{15}{8} - \frac{3}{8y} \\ \implies f_Y(y) &= \boxed{\frac{3}{8y^2}} \end{aligned}$$

Case 2: $1/3 \leq y \leq 1$ corresponding to $1 \leq x \leq 3$

$$\begin{aligned} F_Y(y) &= 1 - \mathbb{P}(X < 1/y) \\ &= 1 - 1/4 \\ &= 3/4 \\ \implies f_Y(y) &= \boxed{0} \end{aligned}$$

Intuitively, since $f_X(x) = 0$ here, $f_Y(y) = \boxed{0}$.

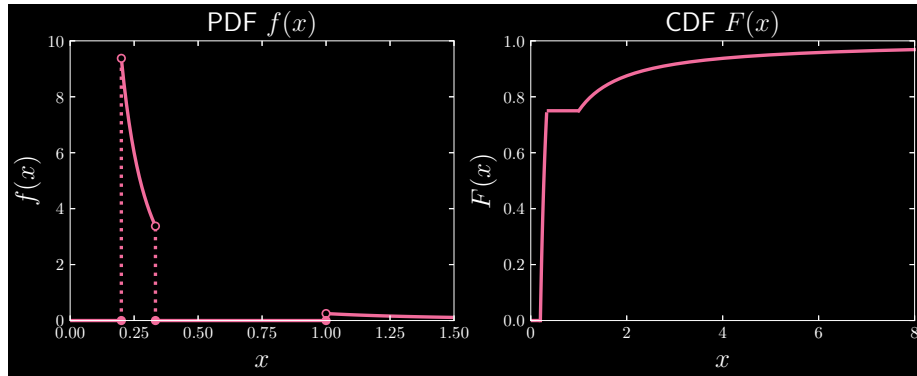
Case 3: $y > 1$ corresponding to $0 < x < 1$

$$\begin{aligned} F_Y(y) &= 1 - \mathbb{P}(X < 1/y) \\ &= 1 - \frac{1}{4y} \\ \implies f_Y(y) &= \boxed{\frac{1}{4y^2}} \end{aligned}$$

Thus the CDF and PDF of Y is given by

$$\begin{aligned} F_Y(y) &= \begin{cases} 0 & y \leq 1/5 \\ 15/8 - 3/8y & 1/5 < y < 1/3 \\ 3/4 & 1/3 \leq y \leq 1 \\ 1 - 1/4y & y > 1 \end{cases} \\ f_Y(y) &= \begin{cases} 3/8y^2 & 1/5 < y < 1/3 \\ 1/4y^2 & y > 1 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Graphing the PDF and CDF:

**Question: 2.14.5**

Let X and Y be discrete random variables. Show that X and Y are independent if and only if $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ for all x and y .

PROOF. ($\implies$) Let $X \sqcup Y$. Then for all x, y , $\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y)$. Since $f_X(x) = \mathbb{P}(X = x)$, $f_Y(y) = \mathbb{P}(Y = y)$, and $f_{X,Y}(x,y) = \mathbb{P}(X = x, Y = y)$, we have

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

($\impliedby$) Let $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ for all x, y . Since $f_{X,Y}(x,y) = \mathbb{P}(X = x, Y = y)$, $f_X(x) = \mathbb{P}(X = x)$ and $f_Y(y) = \mathbb{P}(Y = y)$, we must have

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y)$$

for all x, y . Thus $X \sqcup Y$. □

Question: 2.14.6

Let X have distribution F and density function f and let A be a subset of the real line. Let $I_A(x)$ be the indicator function for A :

$$I_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A. \end{cases}$$

Let $Y = I_A(X)$. Find an expression for the cumulative distribution of Y . (Hint: first find the probability mass function for Y .)

Let $Y = I_A(X)$. Then

$$f_Y(y) = \begin{cases} \mathbb{P}(Y = 1) = \int_A f \, dx & x \in A \\ \mathbb{P}(Y = 0) = 1 - \int_A f \, dx & x \notin A \\ 0 & \text{otherwise} \end{cases}$$

which implies

$$F_Y(y) = \begin{cases} 0 & y < 0 \\ 1 - \int_A f \, dx & 0 \leq y < 1 \\ 1 & y \geq 1 \end{cases}$$

Question: 2.14.7

Let X and Y be independent and suppose that each has a $\text{Uniform}(0, 1)$ distribution. Let $Z = \min\{X, Y\}$. Find the density $f_Z(z)$ for Z . Hint: It might be easier to first find $\mathbb{P}(Z > z)$.

One can follow the hint and observe that $\mathbb{P}(Z > z)$ if and only if

$$\mathbb{P}(X > z)\mathbb{P}(Y > z) = (1 - \mathbb{P}(X < z))(1 - \mathbb{P}(Y < z)) = (1 - F_X(z))(1 - F_Y(z))$$

and applying the fact that $F_X(x), F_Y(y)$ both follow a standard uniform distribution. We will also attempt to derive the density $f_Z(z)$ by modeling the mutual exclusivity of either X or Y being the minimum.

We define Z as

$$Z = \min\{X, Y\} = \begin{cases} X & x < y \\ Y & y < x \end{cases}$$

Since X, Y have standard uniform distribution, it must also be the case that $0 < z < 1$ has non-zero probability, and zero probability otherwise.

Our task is to derive $\mathbb{P}(0 < Z < z)$. Either $X \leq Y$ or $Y \leq X$, and in those cases, we must also have $X \leq z$ and $Y \leq z$, respectively. Thus we derive $F_Z(z)$ on this interval:

$$\begin{aligned} \mathbb{P}(0 < Z \leq z) &= \mathbb{P}(X \leq z)\mathbb{P}(X \leq Y) + \mathbb{P}(Y \leq z)\mathbb{P}(Y \leq X) \\ &= \int_0^z \int_x^1 dy \, dx + \int_0^z \int_y^1 dx \, dy \\ &= \boxed{2z - z^2} \end{aligned}$$

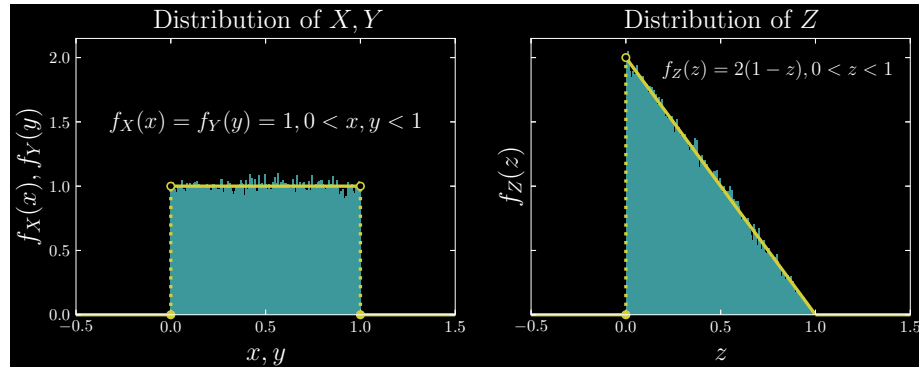
Similarly, we look at $\mathbb{P}(z < Z < 1)$, which corresponds to $1 - F_Z(z)$. As in the previous case, either $X \leq Y$ or $Y \leq X$, but this time these cases correspond to $X > z$ and $Y > z$. We derive

$$\begin{aligned} \mathbb{P}(z < Z < 1) &= \mathbb{P}(X > z)\mathbb{P}(X \leq Y) + \mathbb{P}(Y > z)\mathbb{P}(Y \leq X) \\ &= \int_z^1 \int_x^1 dy \, dx + \int_z^1 \int_y^1 dx \, dy \\ &= \boxed{1 - 2z + z^2} \end{aligned}$$

Either way, we can conclude

$$\begin{aligned} F_Z(z) &= \begin{cases} 0 & z \leq 0 \\ 2z - z^2 & 0 < z < 1 \\ 1 & z \geq 1 \end{cases} \\ f_Z(z) &= \begin{cases} 2(1 - z) & 0 < z < 1 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

And as anticipated, our theoretical work aligns with the empirical:

**Question: 2.14.8**

Let X have CDF F . Find the CDF of $X^+ = \max\{0, X\}$.

Observe that $\mathbb{P}(X^+ = 0) = \mathbb{P}(X \leq 0) = F(0)$. For $X > 0$, we have $\mathbb{P}(X^+ = X) = \mathbb{P}(0 < X \leq x) = F(x) - F(0)$. Then the CDF of X^+ is

$$F_{X^+}(X^+) = \begin{cases} F_X(0) & X \leq 0, X^+ = 0 \\ F_X(x) & X > 0, X^+ = X > 0 \end{cases}$$

Question: 2.14.9

Let $X \sim \text{Exp}(\beta)$. Find $F(x)$ and $F^{-1}(q)$.

$$F(x) = \int_0^x \frac{1}{\beta} \exp(-x/\beta) dx = 1 - \exp(-x/\beta), \quad x > 0$$

Let $q = F(x)$. Then $q = 1 - \exp(-x/\beta)$ implies

$$\begin{aligned} \implies \exp(-x/\beta) &= 1 - q \\ \implies x &= -\beta \ln(1 - q), \quad 0 \leq q < 1 \end{aligned}$$

Question: 2.14.10

Let X and Y be independent. Show that $g(X)$ is independent of $h(Y)$ where g and h are functions.

PROOF. Let $X \sqcup Y$. Consider the joint probability $\mathbb{P}(X, Y)$. By premise, $\mathbb{P}(X, Y) = \mathbb{P}(X)\mathbb{P}(Y)$ for all x, y . Now consider $g(X), h(Y)$. Let

$$\begin{aligned} A &= \{X : g(X) \in A'\} \\ B &= \{Y : h(Y) \in B'\} \end{aligned}$$

where A', B' are the sets containing the respective transformations of $X \in A, Y \in B$. The intuition to have here is that the "link" between $X \in A$ and $g(X) \in A'$ preserves the independence of the former in the latter under transformation.

Then we have

$$\begin{aligned}\mathbb{P}(g(X) \in A') &= \mathbb{P}(X \in A) \\ \mathbb{P}(h(Y) \in B') &= \mathbb{P}(Y \in B) \\ \implies \mathbb{P}(g(X) \in A', h(Y) \in B') &= \mathbb{P}(X \in A, Y \in B)\end{aligned}$$

Applying independence on the last equality,

$$\mathbb{P}(g(X \in A'), h(Y \in B')) = \mathbb{P}(g(X \in A'))\mathbb{P}(h(Y \in B'))$$

ascertaining that $g(X) \sqcup h(X)$. □

Question: 2.14.11

Suppose we toss a coin once and let p be the probability of heads. Let X denote the number of heads and let Y denote the number of tails.

- (a) Prove that X and Y are dependent.
- (b) Let $N \sim \text{Poisson}(\lambda)$ and suppose we toss a coin N times. Let X and Y be the number of heads and tails. Show that X and Y are independent.

PROOF. (a) We have $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(Y = 1) = 1 - p$, where X, Y are the number of heads and tails. Note that $\mathbb{P}(X = 1, Y = 1) = 0$, as we can only have one head or one tail after a single toss. But clearly $\mathbb{P}(X = 1)\mathbb{P}(Y = 1) = p(1 - p) > 0$ for non-zero p , so X, Y are dependent.

(b) The intuition is to decompose the joint probability $f_{X,Y}(x, y)$ into $g(x)h(y)$, where $g(x), h(y)$ are *any* functions (not necessarily densities). We can equivalently find the joint probability that we toss $X = x$ heads *and* toss the coin N times:

$$\begin{aligned}\mathbb{P}(X = x, Y = y) &= \mathbb{P}(X = x, N = x + y) \\ &= \binom{N}{x} p^x (1 - p)^{N-x} e^{-\lambda} \frac{\lambda^N}{N!} \\ &= \binom{x+y}{x} p^x (1 - p)^y e^{-\lambda} \frac{\lambda^{x+y}}{(x+y)!} \\ &= \frac{(x+y)!}{x!y!} p^x (1 - p)^y e^{-\lambda} \frac{\lambda^{x+y}}{(x+y)!} \\ &= \underbrace{\left(e^{-\lambda} \frac{\lambda^x}{x!} p^x \right)}_{g(x)} \underbrace{\left(\frac{\lambda^y}{y!} (1 - p)^y \right)}_{h(y)}\end{aligned}$$

□

Question: 2.14.12

Prove Theorem 2.33.

Theorem (Wasserman 2.33)

Suppose that the range of X and Y is a (possibly infinite) rectangle. If $f(x, y) = g(x)h(y)$ for some functions g and h (not necessarily probability density functions) then X and Y are independent.

PROOF. Let X, Y have density $f(x, y)$, and define

$$A = \{(x, y) : x > 0, y > 0\}$$

Then

$$\begin{aligned} 1 &= \int \int_A f(x, y) \, dx \, dy \\ &= \underbrace{\int_0^\infty g(x) \, dx}_c \underbrace{\int_0^\infty h(y) \, dy}_{1/c} \end{aligned}$$

where $0 < c < 1$. Let $f_X(x) = \frac{1}{c}g(x), x > 0$ and $f_Y(y) = ch(y), y > 0$. Then

$$f(x, y) = g(x)h(y) = \left(\frac{1}{c}g(x)\right)(ch(y)) = f_X(x)f_Y(y)$$

for all x, y . Thus $X \sqcup Y$. □

Question: 2.14.13

Let $X \sim N(0, 1)$ and let $Y = e^X$.

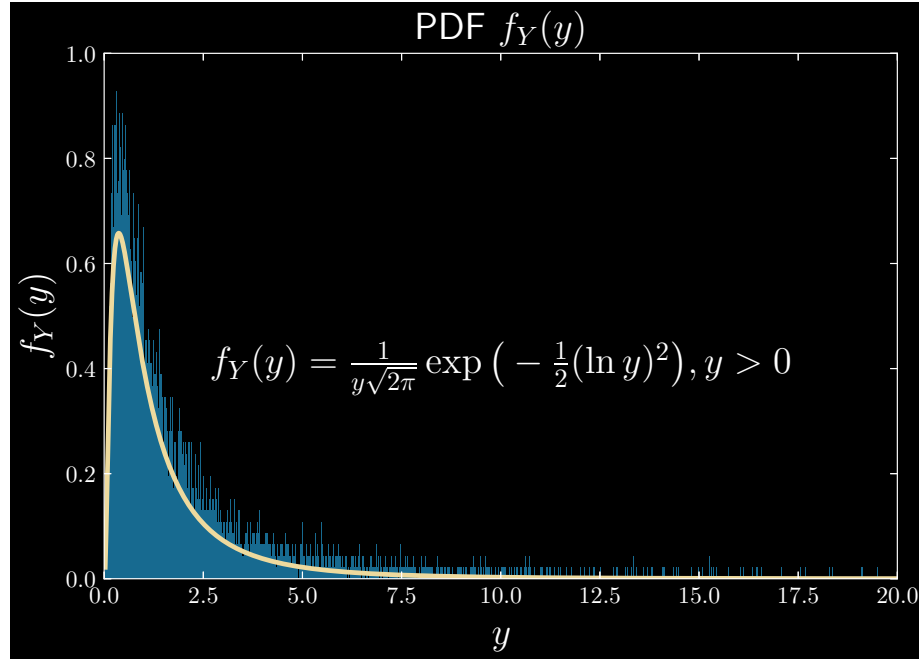
- (a) Find the PDF for Y . Plot it.
- (b) (Computer Experiment.) Generate a vector $x = (x_1, \dots, x_{10,000})$ consisting of 10,000 random standard Normals. Let $y = (y_1, \dots, y_{10,000})$ where $y_i = e^{x_i}$. Draw a histogram of y and compare it to the PDF you found in part (a).

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leq y) \\ &= \mathbb{P}(e^X \leq y) \\ &= \mathbb{P}(X \leq \ln y) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\ln y} \exp\left(-\frac{x^2}{2}\right) \, dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^0 \exp\left(-\frac{x^2}{2}\right) \, dx + \frac{1}{2\pi} \int_0^{\ln y} \exp\left(-\frac{x^2}{2}\right) \, dx \end{aligned}$$

Applying the Leibniz rule, we find

$$F'_Y(y) = f_Y(y) = \frac{1}{y\sqrt{2\pi}} \exp\left(-\frac{1}{2}(\ln y)^2\right), \quad y > 0$$

(b) Our theoretical and empirical results agree:



Question: 2.14.14

Let (X, Y) be uniformly distributed on the unit disk $\{(x, y): x^2 + y^2 \leq 1\}$. Let $R = \sqrt{X^2 + Y^2}$. Find the CDF and PDF of R .

For a joint density involving uniformly distributed variables, $f(x, y) = 1/\text{area}(A)$, where A is the region over which the densities of the constituent variables are defined and non-zero. Since we choose uniformly distributed (x, y) over the unit circle, its area is $A = \pi$.

Integrating in polar coordinates, we derive for $0 < r < 1$:

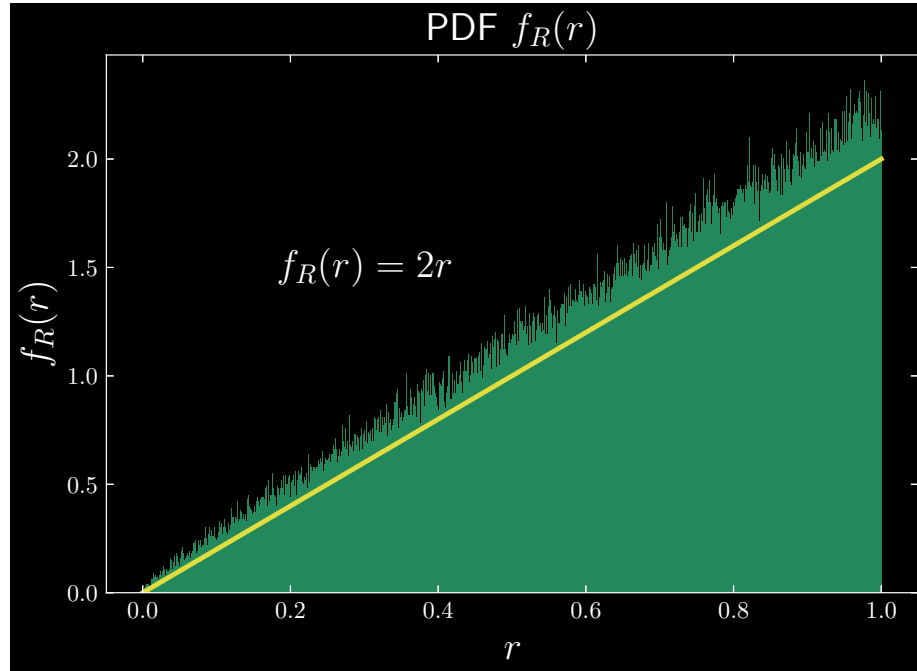
$$F_R(r) = \int_0^{2\pi} \int_0^r \frac{1}{\pi} s \, ds \, d\theta = \int_0^{2\pi} \frac{1}{\pi} \frac{r^2}{2} \, d\theta = r^2$$

giving us the CDF and PDF:

$$F_R(r) = \begin{cases} 0 & r \leq 0 \\ r^2 & 0 < r < 1 \\ 1 & r \geq 1 \end{cases}$$

$$F'_R(r) = f_R(r) = 2r, \quad 0 < r < 1$$

Empirically generating the (x, y) 's agrees with the theoretically derived density:



Question: 2.14.15

(A universal random number generator.) Let X have a continuous, strictly increasing CDF F . Let $Y = F(X)$. Find the density of Y . This is called the probability integral transform. Now let $U \sim \text{Uniform}(0, 1)$ and let $X = F^{-1}(U)$. Show that $X \sim F$. Now write a program that takes $\text{Uniform}(0, 1)$ random variables and generates random variables from an $\text{Exp}(\beta)$ distribution.

PROOF. By assumption, $F_X(x)$ is strictly increasing and continuous. This means $F_X(x)$ is a bijection, and has an inverse $F_X^{-1}(x)$. We use this fact to derive

$$\begin{aligned}
 F_Y(y) &= \mathbb{P}(Y \leq y) \\
 &= \mathbb{P}(F_X(x) \leq y) \\
 &= F_X(F_X^{-1}(y)) \\
 &= y \\
 \implies F_Y^{-1}(y) &= f_Y(y) = 1
 \end{aligned}$$

A shocking result! All probability integral transforms of *any* continuous distribution give us the density to the standard uniform distribution.

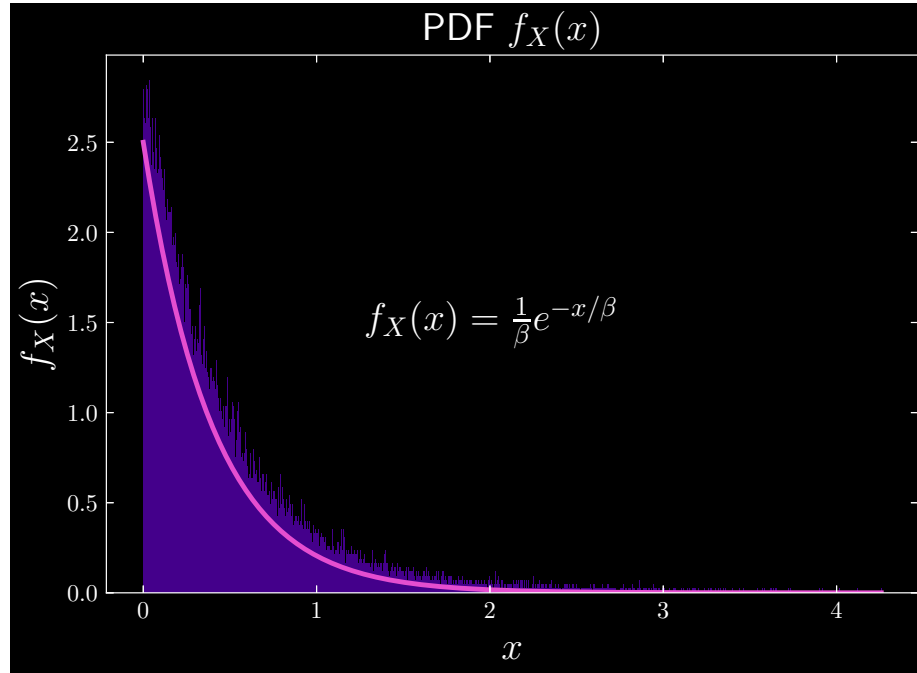
For the next result, let $U \sim \text{Uniform}(0, 1)$ and $X = F^{-1}(U)$. The intuition to have here is that F^{-1} is a quantile function! Again appealing to the bijectivity of F , we have $F(X) = U$. Since $F : X \rightarrow [0, 1]$ and invoking the preservation of

monotonicity under an inverse operation ($x_1 < x_2 \iff F(x_1) < F(x_2)$), we have

$$\mathbb{P}(X \leq x) = \mathbb{P}(F^{-1}(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x)$$

Thus we must have $X \sim F$. $\square$

The quantile function for the $\text{Exp}(\beta)$ distribution is $F^{-1}(q) = -\beta \ln(1 - q)$. Generating the exponential distribution via the probability integral transform:



Question: 2.14.16

Let $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ and assume that X and Y are independent. Show that the distribution of X given that $X + Y = n$ is $\text{Binomial}(n, \pi)$ where $\pi = \lambda/(\lambda + \mu)$.

Hint 1: You may use the following fact: If $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$, and X and Y are independent, then $X + Y \sim \text{Poisson}(\lambda + \mu)$.

Hint 2: Note that $\{X = x, X + Y = n\} = \{X = x, Y = n - x\}$.

PROOF. By definition of the Poisson distribution:

$$f(X + Y = n) = e^{-(\lambda + \mu)} \frac{(\lambda + \mu)^n}{n!}$$

$$f(X = x, Y = n - x) = e^{-\lambda} \frac{\lambda^x}{x!} e^{-\mu} \frac{\mu^{n-x}}{(n-x)!}$$

Proceeding to set up the conditional probability:

$$\begin{aligned}
 f(X | X + Y = n) &= \frac{e^{-\lambda} \frac{\lambda^x}{x!} e^{-\mu} \frac{\mu^{n-x}}{(n-x)!}}{e^{-(\lambda+\mu)} \frac{(\lambda+\mu)^n}{n!}} \\
 &= \frac{\lambda^x \mu^{n-x}}{(\lambda+\mu)^n} \frac{n!}{(n-x)!x!} \\
 &= \binom{n}{x} \left(\frac{\lambda}{\lambda+\mu} \right)^x \left(\frac{\mu}{\lambda+\mu} \right)^{n-x} \\
 &= \binom{n}{x} \left(\frac{\lambda}{\lambda+\mu} \right)^x \left(1 - \frac{\lambda}{\lambda+\mu} \right)^{n-x}
 \end{aligned}$$

ascertaining that $X | X + Y = n \sim \text{Binomial}(n, \pi)$ for $\pi = \lambda/(\lambda + \mu)$. □

Question: 2.14.17

Let

$$f_{X,Y}(x, y) = \begin{cases} c(x + y^2) & 0 \leq x < 1 \text{ and } 0 \leq y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find $\mathbb{P}(X < \frac{1}{2} | Y = \frac{1}{2})$.

First we find the marginal density function of y :

$$f_Y(y) = \int_0^1 c(x + y^2) dx = c(1/2 + y^2)$$

Then the conditional density function is

$$f_{X|Y}(x | y) = \frac{x + y^2}{1/2 + y^2}$$

Now we calculate the desired probability:

$$\begin{aligned}
 \mathbb{P}(X < 1/2 | Y = 1/2) &= \int_0^{1/2} f_{X|Y}(x | 1/2) dx \\
 &= \int_0^{1/2} \frac{x + 1/4}{1/2 + 1/4} dx \\
 &= \frac{1/8 + 1/8}{1/2 + 1/4} \\
 &= \boxed{1/3}
 \end{aligned}$$

Question: 2.14.18

Let $X \sim N(3, 16)$. Solve the following using the Normal table and using a computer package.

- (a) Find $\mathbb{P}(X < 7)$.
- (b) Find $\mathbb{P}(X > -2)$.
- (c) Find x such that $\mathbb{P}(X > x) = .05$.
- (d) Find $\mathbb{P}(0 \leq X < 4)$.
- (e) Find x such that $\mathbb{P}(|X| > |x|) = .05$.

It is best to use `scipy.stats` here:

```
from scipy.stats import norm

normal_dist = norm(3, 4)

# (a)
prob_a = normal_dist.cdf(7)
print("The probability for (a) is", prob_a)

# (b)
prob_b = 1 - normal_dist.cdf(-2)
print("The probability for (b) is", prob_b)

# (c)
x_c = normal_dist.ppf(0.95)
print("The value of x for (c) is", x_c)

# (d)
prob_d = normal_dist.cdf(4) - normal_dist.cdf(0)
print("The probability for (d) is", prob_d)

# (e)
x = np.arange(9.5, 9.7, 0.00000001)
array = np.round(normal_dist.cdf(x) - normal_dist.cdf(-x), decimals=9)
index = np.where(array == 0.95)
x_e = x[index[0]]
print("Some values of x for (e) are", x_e)
```

Using the above calculations:

- (a) $\mathbb{P}(X < 7) = 0.841$
- (b) $\mathbb{P}(X > -2) = 0.9894$
- (c) $x = 9.579$
- (d) $\mathbb{P}(0 \leq X < 4) = 0.372$

(e) $x = 9.611$

For (e), the intuition behind the absolute values is to evaluate the probabilities at the tails of the distribution. Given $|x| > 0$, we can equivalently write

$$\begin{aligned}\mathbb{P}(|X| > |x|) &= \mathbb{P}(X > |x|) + \mathbb{P}(X < -|x|) \\ &= 1 - \mathbb{P}(X \leq |x|) + \mathbb{P}(X < -|x|) = 0.05 \\ \implies 0.95 &= \mathbb{P}(X \leq |x|) - \mathbb{P}(X < -|x|)\end{aligned}$$

Numerically iterating enables us to identify the value of $|x| = x$ without needing to convert to a standard normal distribution.

Question: 2.14.19

Prove formula (2.12).

Property (Wasserman Formula 2.12)

For a strictly monotone increasing or strictly monotone decreasing function $r(X)$ with an inverse $s = r^{-1}$, we have

$$f_Y(y) = f_X(s(y)) \left| \frac{ds(y)}{dy} \right|.$$

PROOF. Case 1: Increasing $r(X)$. Derive

$$\begin{aligned}F_Y(y) &= \mathbb{P}(Y \leq y) = \mathbb{P}(r(X) \leq y) \\ &= \mathbb{P}(X \leq r^{-1}(y)) \\ &= \mathbb{P}(X \leq s(y)) \\ &= \int_{-\infty}^{s(y)} f_X(x) dx\end{aligned}$$

Implying $F'_Y(y) = f_Y(y) = f_X(s(y)) \frac{ds(y)}{dy}$. Note that since $r(X)$ is strictly monotone increasing by premise, $ds(y)/dy > 0$.

Case 2: Decreasing $r(X)$. Derive

$$\begin{aligned}F_Y(y) &= \mathbb{P}(Y \leq y) = \mathbb{P}(r(X) \leq y) \\ &= \mathbb{P}(X \geq r^{-1}(y)) \\ &= \mathbb{P}(X \geq s(y)) \\ &= 1 - \mathbb{P}(X \leq s(y)) \\ &= 1 - \int_{-\infty}^{s(y)} f_X(x) dx \\ F'_Y(y) &= f_Y(y) = -f_X(s(y)) \frac{ds(y)}{dy}\end{aligned}$$

where the second equality follows because $r(X)$ is strictly monotonically *decreasing*. That same fact also allows us to say that $ds(y)/dy < 0$. To encapsulate both the increasing and decreasing cases, we have

$$f_Y(y) = f_X(s(y)) \left| \frac{ds(y)}{dy} \right|$$

□

Question: 2.14.20

Let $X, Y \sim \text{Uniform}(0, 1)$ be independent. Find the PDF for $X - Y$ and X/Y .

The joint probability density function is given by

$$f_{X,Y}(x, y) = \begin{cases} 1 & 0 < x, y < 1 \\ 0 & \text{otherwise} \end{cases}$$

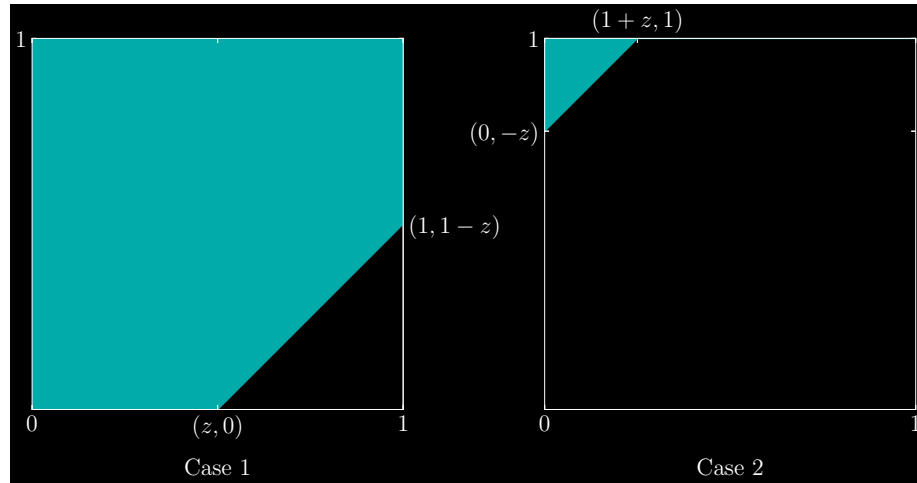
$X - Y$

Illustrations are particularly useful here to understand where the piecewise segments of $f_Z(z)$ are defined.

Given $Z = X - Y$, it must be the case that either $0 < z < 1$ or $-1 < z < 0$. To find A_z , the support for the probability density function, we find all x, y satisfying

$$A_z = \{ (x, y) : x - y \leq z \}$$

Now, there is a nice geometric intuition in that the piecewise CDF is given by the area of the support on each interval $0 < z < 1, -1 < z < 0$. Graphically, this support in the first and second cases is:



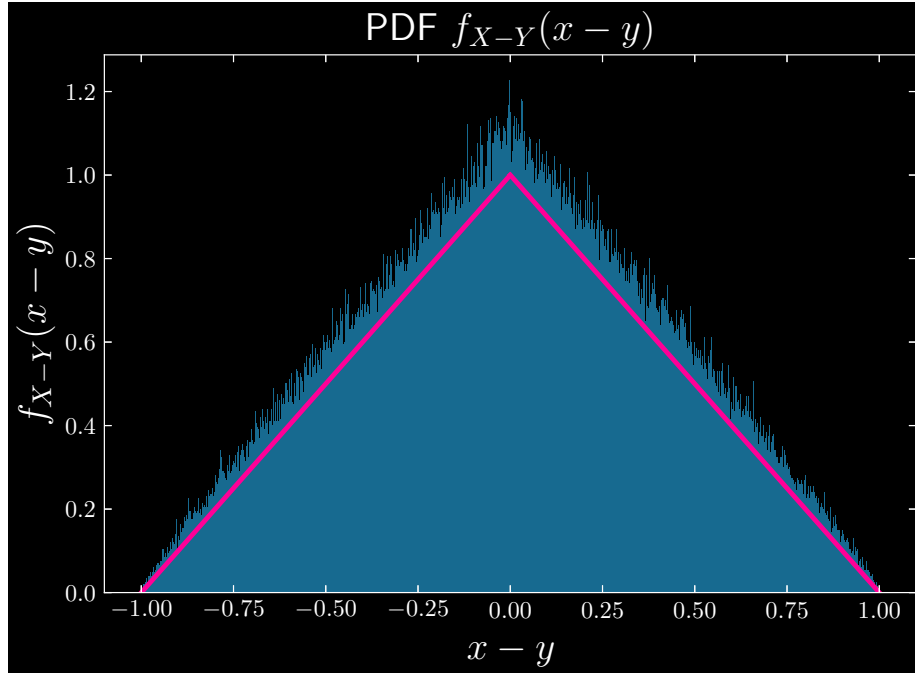
Calculating the area of the shaded parts, we can derive the CDF as

$$F_Z(z) = \begin{cases} 0 & z < -1 \\ \frac{1}{2}(1+z)^2 & -1 < z < 0 \\ 1 - \frac{1}{2}(1-z)^2 & 0 < z < 1 \\ 1 & z > 1 \end{cases}$$

and consequently, the PDF is

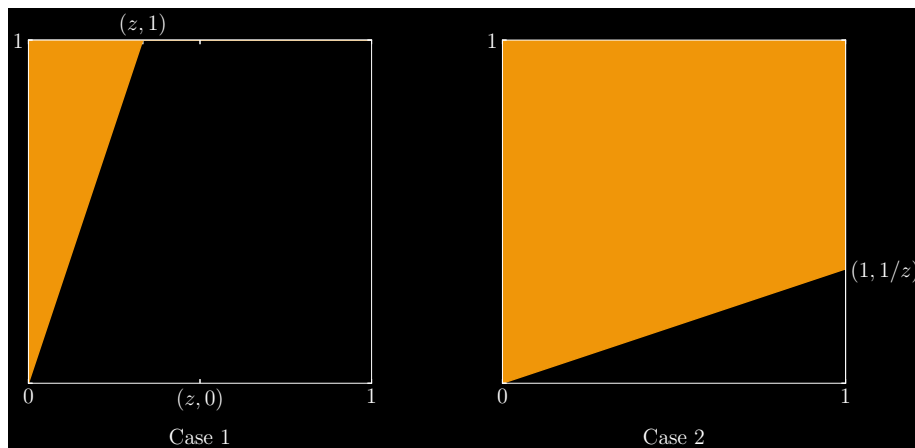
$$f_Z(z) = \begin{cases} 1+z & -1 < z < 0 \\ 1-z & 0 < z < 1 \\ 0 & \text{otherwise} \end{cases}$$

which agrees with our empirical simulation:



X/Y

Once again finding the supports for $Z = X/Y$, observe that the intervals we examine are $0 < z < 1$ and $z > 1$. The areas are graphed as



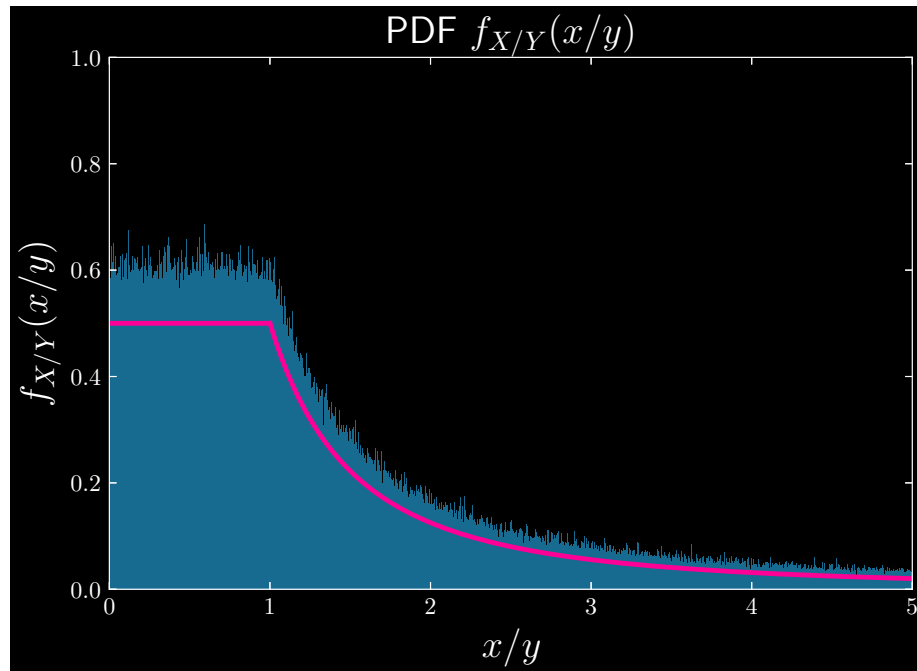
with CDF

$$F_Z(z) = \begin{cases} 0 & z < 0 \\ \frac{z}{2} & 0 < z < 1 \\ 1 - \frac{1}{2z} & z > 1 \end{cases}$$

and following that, the PDF

$$f_Z(z) = \begin{cases} \frac{1}{2} & 0 < z < 1 \\ \frac{1}{2z^2} & z > 1 \\ 0 & \text{otherwise} \end{cases}$$

Once again, the theoretical and the empirical align:



Question: 2.14.21

Let $X_1, \dots, X_n \sim \text{Exp}(\beta)$ be IID. Let $Y = \max\{X_1, \dots, X_n\}$. Find the PDF of Y . Hint: $Y \leq y$ if and only if $X_i \leq y$ for $i = 1, \dots, n$.

Taking advantage of the hint and the IID random variables, observe that $\mathbb{P}(Y \leq y) = \prod_{i=1}^n \mathbb{P}(X_i \leq y)$. The derivation is straightforward:

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leq y) = \prod_{i=1}^n \int_0^y \frac{1}{\beta} e^{-x_i/\beta} dx_i \\ &= \prod_{i=1}^n (1 - e^{-y/\beta}) \\ &= (1 - e^{-y/\beta})^n \\ \implies F'_Y(y) &= f_Y(y) = \frac{d}{dy} (1 - e^{-y/\beta})^n = \boxed{\frac{n}{\beta} e^{-y/\beta} (1 - e^{-y/\beta})^{n-1}} \end{aligned}$$